```
import pandas as pd
import numpy as np
```

```
df=pd.read_csv("healthcare_basic_dataset.csv")
```

```
df.head()
```

	Patient_ID	Age	Gender	BMI	Blood_Pressure	Diabetes
0	1	25	Male	22.5	120	No
1	2	34	Female	24.1	118	No
2	3	45	Male	26.3	130	Yes
3	4	29	Female	23.8	115	No
4	5	50	Male	27.5	135	Yes

```
df.shape
```

```
(20, 6)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Patient_ID      20 non-null     int64
 1   Age             20 non-null     int64
 2   Gender          20 non-null     object
 3   BMI             20 non-null     float64
 4   Blood_Pressure  20 non-null     int64
 5   Diabetes        20 non-null     object
dtypes: float64(1), int64(3), object(2)
memory usage: 1.1+ KB
```

```
df.isnull().sum()
```

	0
Patient_ID	0
Age	0
Gender	0
BMI	0
Blood_Pressure	0
Diabetes	0

dtype: int64

```
df=pd.read_csv("student_performance_100.csv")
print(df)
```

```
    Student_ID   Name  Age Gender    Subject  Marks  Attendance Result
0          101   Amit   19      M       Math   78.0        85.0   Pass
1          102   Neha   20      F    Physics   88.0        92.0   Pass
2          103  Rahul   19      M  Chemistry   65.0        70.0   Pass
3          104  Pooja   20      F       Math   90.0        95.0   Pass
4          105  Ankit   21      M    Physics    NaN        60.0   Fail
..         ...    ...  ...    ...        ...    ...         ...    ...
95         196  Sneha   19      F  Chemistry   72.0        80.0   Pass
96         197  Rohit   20      M       Math   55.0        65.0   Fail
97         198  Kavya   19      F    Physics   82.0         NaN   Pass
98         199   Aman   21      M  Chemistry   60.0        75.0   Pass
99         200   Neha   20      F       Math   88.0        92.0   Pass

[100 rows x 8 columns]
```

```
df.isnull().sum()
```

	0
Student_ID	0
Name	0
Age	0
Gender	0
Subject	0
Marks	10
Attendance	10
Result	0

dtype: int64

```
df.shape
```

```
(100, 8)
```

```
df.columns
```

```
Index(['Student_ID', 'Name', 'Age', 'Gender', 'Subject', 'Marks', 'Attendance',
       'Result'],
      dtype='object')
```

```
df['Gender'].value_counts()['F']
```

```
np.int64(50)
```

```
df['Marks']
```

	Marks
0	78.0
1	88.0
2	65.0
3	90.0
4	NaN
...	...
95	72.0
96	55.0
97	82.0
98	60.0
99	88.0

100 rows × 1 columns

dtype: float64

display all those rows and column having nan values

```
rows_with_nulls=df[df.isnull().any(axis=1)]
rows_with_nulls
```

	Student_ID	Name	Age	Gender	Subject	Marks	Attendance	Result
4	105	Ankit	21	M	Physics	NaN	60.0	Fail
7	108	Kavya	19	F	Physics	82.0	NaN	Pass
14	115	Ankit	21	M	Physics	NaN	60.0	Fail
17	118	Kavya	19	F	Physics	82.0	NaN	Pass
24	125	Ankit	21	M	Physics	NaN	60.0	Fail
27	128	Kavya	19	F	Physics	82.0	NaN	Pass
34	135	Ankit	21	M	Physics	NaN	60.0	Fail
37	138	Kavya	19	F	Physics	82.0	NaN	Pass
44	145	Ankit	21	M	Physics	NaN	60.0	Fail
47	148	Kavya	19	F	Physics	82.0	NaN	Pass
54	155	Ankit	21	M	Physics	NaN	60.0	Fail
57	158	Kavya	19	F	Physics	82.0	NaN	Pass
67	168	Kavya	19	F	Physics	82.0	NaN	Pass
74	175	Ankit	21	M	Physics	NaN	60.0	Fail
94	195	Ankit	21	M	Physics	NaN	60.0	Fail
97	198	Kavya	19	F	Physics	82.0	NaN	Pass

```
null_rows_count=df[df.isnull().any(axis=1)].shape
null_rows_count
```

```
(20, 8)
```

```
arr1=np.array([
    [1,2,3,4],[5,6,7,8]
])
print(arr1)
```

```
[[1 2 3 4]
 [5 6 7 8]]
```

```
print(arr1[1,1])
```

```
6
```

```
sub=arr1[0:2,0:1]
print(sub)
```

```
[[1]
 [5]]
```

Start coding or generate with AI.